Warsaw Real Estate AVM

An Econometric Automated Valuation Engine with Spatial GIS Pipeline

Marcel Zenderowski
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Abstract

This document outlines the technical architecture, econometric formulation, and empirical performance of the Warsaw Real Estate Automated Valuation Model (AVM). Built on an empirical sample of $5,471$ residential apartment listings, the engine integrates spherical Haversine geodesic metrics to transit networks (Warsaw Metro M1/M2), spatial k-NN infrastructure imputations, and regularized ensemble decision trees to deliver robust fair-market property valuations within an interactive analytical terminal.

Project Metadata:

- **Stack:** Python, Scikit-Learn, Dash, Plotly, Pandas, NumPy, GIS (Haversine Distance).
- **Core Metrics:** Out-of-Sample $R^2 \approx 0.8510$, MAE $\approx$ 107,500 PLN, MAPE $\approx 13.4\%$.
- **Status:** Complete & Production-Ready Interactive Terminal.

1 Executive Summary & Core Mandate

The Warsaw Real Estate AVM is designed to estimate the fair market value ($\hat{y}_i$) of residential properties across all districts of Warsaw based on spatial coordinates, physical attributes, and building characteristics. The pipeline circumvents classical spatial estimation limitations by calculating exact geodesic distances to transit hubs (Metro M1/M2) and PKiN (Centrum), combined with a high-depth Random Forest Regressor and a standalone interactive Dash interface.

2 Dataset & Descriptive Diagnostics

The model was trained and evaluated on N = 5,471 property listings across Warsaw. Summary statistics for the primary econometric variables are presented in Table 1.

Table 1: Descriptive Statistics of the Warsaw Housing Dataset (N = 5,471)

Metric	Price (PLN)	Area (m^2)	Rooms	Floor	Build Year	Centrum Dist. (km)
Mean	919,840	60.86	2.72	3.78	1989.05	6.17
Std. Dev.	407,435	22.80	0.93	3.04	28.69	3.32
Min	327,635	25.00	1.00	1.00	1851.00	0.02
25% (Q1)	630,000	45.02	2.00	2.00	1970.00	3.71
50% (Med)	799,000	56.60	3.00	3.00	1998.00	6.05
75% (Q3)	1,100,000	72.32	3.00	5.00	2013.00	8.29
Max	2,500,000	150.00	6.00	29.00	2023.00	16.94

3 Spatial Engineering & Mathematics

3.1 Spherical Geodesy (Haversine Formulation)

The great-circle distance d between property coordinates (ϕ_1, λ_1) and target transit/central hubs (ϕ_2, λ_2) is resolved exogenously using the spherical Haversine formula:

$$d = 2R \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)}\right) \tag{1}$$

where $R = 6371.0$ km, $\Delta\phi = \phi_2 - \phi_1$, and $\Delta\lambda = \lambda_2 - \lambda_1$.

3.2 Feature Vector Assembly

- `dist_to_metro_km`: Minimum orthogonal distance to the nearest of 37 active Metro stations (Lines M1 and M2).
- `centreDistance`: Direct geodesic distance to the Central Business District (PKiN: 52.2297° N, 21.0122° E).
- `avg_infra_distance`: Composite arithmetic mean across local amenities (kindergartens, schools, clinics, pharmacies).
- `floor_ratio`: Vertical placement index relative to the total structure height ($\frac{\text{floor}}{\text{floorCount}}$).
- **Spatial k-NN Imputer**: Dynamically fills missing point-of-interest indicators during single-point valuation queries using a distance-weighted k-Nearest Neighbors regressor ($k = 5$).

4 Model Architecture & Empirical Performance

The core valuation engine utilizes a `RandomForestRegressor` ensemble trained on an 80/20 train-test out-of-sample holdout partition.

Table 2: Model Specification and Out-of-Sample Diagnostics

Parameter / Metric	**Value**
Ensemble Architecture	Random Forest Regressor (`n_estimators=180, max_depth=18`)
Evaluation Split	80% Training ($N_{tr} = 4{,}376$) / 20% Holdout ($N_{te} = 1{,}095$)
Coefficient of Determination (R^2)	**0.8510**
Mean Absolute Error (MAE)	**107,508** PLN
Root Mean Squared Error (RMSE)	**152,000** PLN
Mean Absolute Percentage Error (MAPE)	**13.4**% (Relative to Median Price)

4.1 Feature Importance Decomposition

Gini impurity reduction metrics demonstrate the structural hierarchy of price determinants:

1. **Apartment Area (`squareMeters`):** Dominant volumetric driver ($> 60\%$ total weight).
2. **Spatial Transit Access (`centreDistance` + `dist_to_metro_km`):** Combined $> 16\%$ weight.
3. **Physical Attributes (`buildYear`, `floor`, `hasElevator`):** Local quality modifiers.

5 Terminal Architecture (`app.py`)

The production application provides four core diagnostic and valuation views:

1. **Overview & Analytics:** Vector Mapbox visualizer with M1/M2 metro line overlays and bivariate density heatmaps.

2. **Geo-Spatial Valuation Engine:** Real-time pricing terminal with automated district coordinate presets and dynamic inference.
3. **Out-of-Sample R^2 Inspector:** Live ablation module tracking generalization performance across arbitrary feature subsets.
4. **Model Feature Weights:** Gini importance horizontal decomposition.